

Lab 4.1 (Linear mixed model)

The dataset `data_growth.csv` contains the heights of 197 children suffering from acute lymphoblastic leukaemia, and receiving three different treatments.

- a) Load the dataset `data_growth.csv` into R, and plot the data for three selected children on the same graphic. Comment it.
- b) Plot the full data on three panels, one panel for each treatment type. Comments.
- c) Fit a linear mixed model with random intercept to explain height in terms of age and treatment. Comments.
- d) Fit a linear mixed model with random slope to explain height in terms of age and treatment. Comments.
- e) Fit a linear mixed model with random intercept and random slope to explain height in terms of age and treatment. Plot the predicted random effects with confidence intervals. Comments.
- f) Compare model in e) versus that in c).